

Right Anodal tDCS in Chronic Aphasia: Effects on Attentional Networks and Language Recovery

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Background: Attention deficits are common in post-stroke aphasia and may constrain rehabilitation outcomes, but the effects of tDCS on specific attentional networks remain unclear. **Objective:** To explore whether right-hemisphere anodal tDCS combined with Intensive Language Therapy and Donepezil (ILAT- DNP) modulates alerting, orienting, and executive control, and whether attentional changes relate to language recovery. **Method:** Twenty adults with chronic post-stroke aphasia receiving 10 mg/day donepezil were assigned to anodal tDCS + ILAT- DNP (n=10) or sham tDCS + ILAT- DNP (n=10). Attention was assessed with the Attention Network Test for Interactions (ANTI). Language, communication, and quality of life were assessed with the WAB-AQ, CAL, and SAQOL-39. Pre-post changes were analysed using t-tests or Mann-Whitney U tests, linear mixed models controlling for age, lesion volume, and time post-stroke, and moderation analyses. **Results:** Both groups improved from baseline to post-treatment in language (WAB-AQ: $\beta=3.70$, $p=.004$), communication (CAL: $p=.020$), and quality of life (SAQOL-39: $p=.002$), indicating a therapy effect independent of stimulation. No between-group differences emerged for orienting, executive control, or clinical outcomes. The only between-group difference was in alerting, defined as auditory-cue benefit (RT no tone – RT tone): the anodal group increased alerting ($\Delta=+10.5$ ms), whereas sham declined ($\Delta=-25.7$ ms; $d=1.07$, $p=.041$; LMM interaction $\beta=-35.22$, $p=.025$). Alerting change moderated WAB-AQ gains by group ($\beta=-0.125$, $p=.001$, $R^2=.78$): in sham, greater alerting loss was associated with larger WAB-AQ gains, whereas this relationship was absent under anodal tDCS. **Conclusion:** Anodal tDCS did not enhance language recovery beyond ILAT, but may selectively preserve auditory alerting and alter its coupling with language gains.